Cedric Richter

Postdoctoral Researcher at Humboldt-Universität zu Berlin

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Research Interests

My research focuses on tools and techniques at the intersection of **Machine Learning**, **Software Engineering**, and **Formal Methods**. I developed techniques for finding bugs by learning from previous developer's mistakes, for adapting and combining software verification tools, and for cooperations between AI and formal verification. More broadly, I am interested in how advances in AI can support software developers in building reliable software systems.

Education

2021 – 2025 **Carl von Ossietzky Universität Oldenburg**, Oldenburg
Ph.D. in computer science, Grade: *summa cum laude*.
Advisor: Heike Wehrheim
Topic: Neural Bug Detection
Disputation: 16.05.2025

2017 – 2019 **Paderborn University**, Paderborn
M.Sc in computer science, with distinction.

Topic: Neural Algorithm Selection for Software Verification
Date: 09 / 2019
Thesis Supervisor: Heike Wehrheim
Grade: 1,2

2014 - 2017 **Paderborn University**, Paderborn
B.Sc in computer science.

Topic: Evolutionary Strategies for Timed Game Automata
Date: 10 / 2017
Thesis Supervisor: Heike Wehrheim
Grade: 1,4

Positions and Experience

Oct 2026 – present
Humboldt-Universität zu Berlin, Berlin
Postdoctoral Researcher in the Software Engineering Group led by Lars Grunske

Oct 2025 – Sep 2026
University of Luxembourg, Luxembourg
Interdisciplinary Centre for Security, Reliability and Trust (SnT)
Postdoctoral Researcher in the SerVal Group led by Mike Papadakis

April 2021 – Sep 2025
Carl von Ossietzky Universität Oldenburg, Oldenburg
Research assistant in the Formal Methods Group led by Heike Wehrheim

Oct 2019 – Mar 2021
Paderborn University, Paderborn
Research assistant in the Formal Methods Group led by Heike Wehrheim

Oct 2016 - Sep 2019
Paderborn University, Paderborn
Student assistant in the Formal Methods Group led by Heike Wehrheim

April 2015 – Sep 2016
Paderborn University, Paderborn
Teaching assistant for undergraduate lectures, Basic Programming

Research funding

2026 – present
DFG Walter Benjamin Programme
SDECOPILOT: Assisting Developers in Building Correct Software with Debugging Agents
Humboldt-Universität zu Berlin

Research project investigating the combination of AI-based techniques and established software engineering methods for automated bug discovery and program repair.

Peer-reviewed Conference and Journal Publications

ICSE'25
Cooperative Software Verification via Dynamic Program Splitting
Cedric Richter, Marek Chalupa, Marie-Christine Jakobs, and Heike Wehrheim

47th International Conference on Software Engineering

FMSD'25 **Construction of verifier combinations from off-the-shelf components**
Dirk Beyer, Sudeep Kanav, Tobias Kleinert, and **Cedric Richter**

Formal Methods in System Design (Journal)

FASE'24 **Can ChatGPT support software verification?**
Christian Janßen, **Cedric Richter**, and Heike Wehrheim

International Conference on Fundamental Approaches to Software Engineering

TACAS'24 **BubaaK-SpLit: Split what you cannot verify**
Marek Chalupa and **Cedric Richter**

30th International Conference on Tools and Algorithms for the Construction and Analysis of Systems

SCP'24 **Ranged Program Analysis on Path Ranges**
Jan Haltermann, Marie-Christine Jakobs, **Cedric Richter**, and Heike Wehrheim

Science of Computer Programming (Journal)

ASE'23 **How to Train Your Neural Bug Detector: Artificial vs Real Bugs**
Cedric Richter and Heike Wehrheim

38th International Conference on Automated Software Engineering

SEFM'23 **Robustness Testing of Software Verifiers**
Florian Dyck, **Cedric Richter**, and Heike Wehrheim

International Conference on Software Engineering and Formal Methods

SEFM'23 **Ranged Program Analysis via Instrumentation**
Jan Haltermann, Marie-Christine Jakobs, **Cedric Richter,** and Heike Wehrheim

International Conference on Software Engineering and Formal Methods

FASE'23 **Parallel Program Analysis via Range Splitting**
Jan Haltermann, Marie-Christine Jakobs, **Cedric Richter**, and Heike Wehrheim

26th International Conference on Fundamental Approaches to Software Engineering

ASE'22 **Are Neural Bug Detectors Comparable to Software Developers on Variable Misuse Bugs?**
Cedric Richter, Jan Haltermann, Marie-Christine Jakobs, Felix Pauck, Stefan Schott, and Heike Wehrheim

37th IEEE/ACM International Conference on Automated Software Engineering

ICST'22 **Learning Realistic Mutations: Bug Creation for Neural Bug Detectors**
Cedric Richter and Heike Wehrheim

15th IEEE Conference on Software Testing, Verification and Validation

MSR'22 **TSSB-3M: Mining single statement bugs at massive scale**
Cedric Richter and Heike Wehrheim

19th International Conference on Mining Software Repositories

FASE'22 **Construction of Verifier Combinations Based on Off-the-Shelf Verifiers**
Dirk Beyer, Sudeep Kanav, and **Cedric Richter**

25th International Conference on Fundamental Approaches to Software Engineering

FASE'21 **CoVeriTest with Adaptive Time Scheduling**
Marie-Christine Jakobs and **Cedric Richter**

24th International Conference on Fundamental Approaches to Software Engineering

ASE'20 **Attend and Represent:**
A Novel View on Algorithm Selection for Software Verification
Cedric Richter and Heike Wehrheim

35th IEEE/ACM International Conference on Automated Software Engineering

JASE'20 **Algorithm Selection for Software Validation Based on Graph Kernels**

Cedric Richter, Eyke Hüllermeier, Marie-Christine Jakobs, and Heike Wehrheim

Journal of Automated Software Engineering 2020

TACAS'19 **PeSCo: Predicting Sequential Combinations of Verifiers**
Cedric Richter and Heike Wehrheim

25th International Conference on Tools and Algorithms for the Construction and Analysis of Systems

Teaching Experience

Teaching Assistant

Teaching assistantships involve preparing and presenting exercises, coordination of student assistants, preparing and grading exams, and organizing office hours.

Winter 2023 **Basics of Theoretical Computer Science**
Bachelor course – Languages, Turing machines, Complexity

Summer 2023 **Formal Methods**
Bachelor course – Petri Nets, CCS, Timed Automata, Z

Winter 2022 **Basics of Theoretical Computer Science**
Bachelor course – Languages, Turing machines, Complexity

Summer 2022 **Model Checking**
Master course – Modeling, LTL, CTL, Bisimulation

Winter 2021 **Practical Software Verification**
Master course – Development of custom Verification Technique

Summer 2021 **Software Analysis**
Master course – Dataflow Analysis, Abstract Interpretation, Slicing

Winter **Programming**

2020 Bachelor course – Programming in Python

Summer 2020 **Modeling Software with Formal Methods**
Bachelor course – Petri Nets, CCS, Timed Automata, Z

Winter 2019 **Software Analysis**
Master course – Dataflow Analysis, Abstract Interpretation, Slicing

Mentor

Mentorships involve organizing regular office hours and support in seminar theses and presentations.

Winter 2024 **Logical Methods in AI Verification seminar**
Master seminar

Winter 2023 **Logical Methods in AI Verification seminar**
Master seminar

Winter 2020 **Advanced Topics in Software Analysis seminar**
Master seminar

Summer 2020 **Advanced Techniques in Software Analysis seminar**
Master seminar

Winter 2019 **Software Testing seminar**
Master seminar

Advising and Mentoring

Master thesis — Melchisedek Hengue Tuomo. *Residual Program Generation For Ranged Symbolic Execution. 2021*

Bachelor thesis — Christian Janßen. *Generating loop invariants for program verification with ChatGPT. 2023*
See paper at FASE'24

Darwin Kaufmann. *Evaluierung von ChatGPT im Kontext der Software Verifikation. 2023*

Florian Dyck. *Benchmarkgenerierung für Verifikationswerkzeuge mittels semantikerhaltender Transformation. 2022*
See paper at SEFM'23

Reviewing and Service

Program commitees	ACM International Conference of Software Engineering (FSE), Research Track, 2027
	IEEE/ACM International Conference on Automated Software Engineering (ASE), Research Track, 2026
	ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), Research Track, 2026
	IEEE/ACM International Conference on Automated Software Engineering (ASE), Research Track, 2025
	ACM International Conference of Software Engineering (FSE), Tool Demonstration Track, 2025
	International Conference on Software Engineering (ICSE), ShadowPC, 2025
	IEEE Security Development Conference (SecDev), 2024
	Competition on Software Verification (SV-COMP). Jury member, 2018 – 2023
Journal reviewer	Empirical Software Engineering (EMSE), 2026
	Science of Computer Programming (SICP), 2026
	Formal Methods in System Design (FORM), 2023
Artifact Evaluation commitees	International Conference on Software Engineering (ICSE), 2024
	International Conference on integrated Formal Methods (iFM), 2023
	International Conference on Formal Methods in Software Engineering (FormaliSE), 2023
	International Conference on Computer Aided Verification (CAV), 2022
	International Conference on Fundamental Approaches to Software Engineering (FASE), 2022

International Conference on integrated Formal Methods (iFM), 2022

ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), 2021

European Conference on Object-Oriented Programming (ECOOP), 2020

International Conference on Tests and Proofs (TAP), 2019